Trainings

Select Service Tools

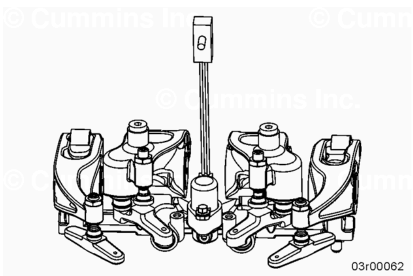
Recommended Cummins® Service Tools

- Engine barring tool, Part Number 4919092
- Camshaft timing pin, Part Number 5572844

Additional Service Items

- Feeler gauge with elbow.

General Information



Overhead setting **must** be performed at the interval specified in the operation and maintenance manual, owners manual, or when engine repairs cause removal of the rocker levers and/or loosening of the adjusting screws.

The brakes are designed as integrated rocker levers with the exhaust rockers.

Excessive valve lash prior to this can indicate an overhead set incorrectly from a previous repair, worn valve stems, crossheads, camshaft, or rocker levers.

Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-009.html)

Preparatory Steps

⚠ WARNING ⚠

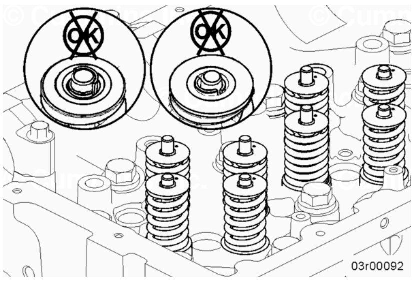
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html)

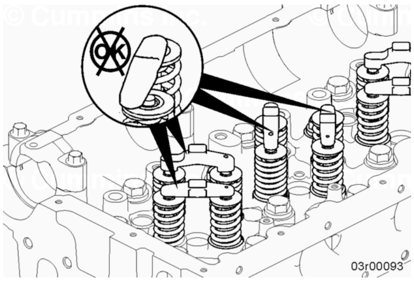
Inspect

The collets **must** be seated properly on each valve stem. Visually inspect the collets of each valve before installing the crosshead or adjusting the overhead.

If needed, adjust the collet. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)



The pocket of the crosshead **must** be seated over the valve stem. Visually inspect the crosshead to ensure it is properly installed.



Measure

Follow the Adjust section in this procedure to locate correct cylinder to measure valve lash.

Use a set of feeler gauges to measure the amount of clearance (lash) between the crosshead and the rocker lever foot.

Measure and record the intake and exhaust valve lash. If the valve lash is **not** within the specifications listed below, the valve **must** be adjusted. See the Adjust section in this procedure.

Lash Check Limits			
	mm		in
Intake	0.30	MIN	0.012

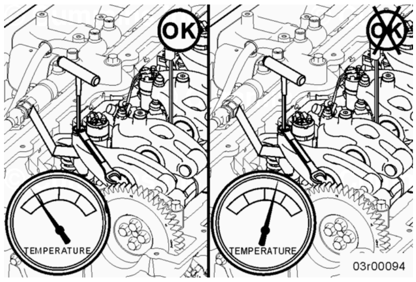
	0.50	MAX	0.020
Exhaust	0.65	MIN	0.026
	0.96	MAX	0.038

Note : Checking the overhead setting is usually performed as part of a troubleshooting procedure. If the lash measurement is out of specification, adjust the lash to the nominal specification.

Note : Lash check limits can not be used as the justification to adjust valve lash or not in scheduled maintenance. The valve lash must be reset to nominal valve when maintenance interval occurs.

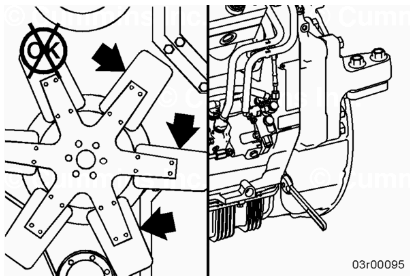
Adjust

All overhead adjustments **must** be made when the engine is cold (any stabilized coolant temperature at 60°C [140°F] or below).



⚠ WARNING ⚠

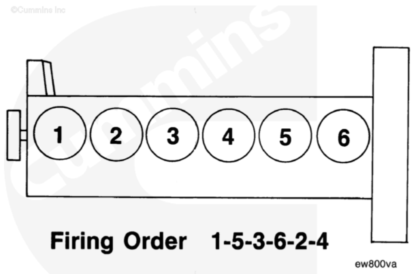
Do not pull or pry on the fan to manually rotate the engine. To do so can damage the fan blades. Damaged fan blades can cause premature fan failures which can result in serious personal injury or property damage.



Remove the barring adapter access plug from the bottom of the flywheel housing.

The cylinders are numbered from the front of the engine (1-2-3-4-5-6).

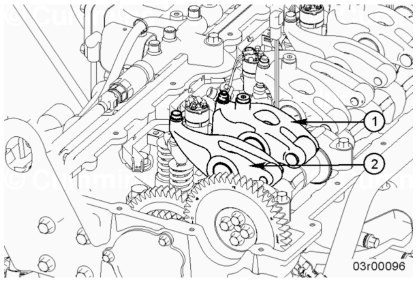
The engine firing order is 1-5-3-6-2-4.



Each cylinder has two rocker levers:

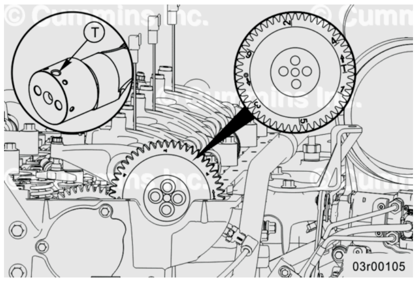
- Exhaust rocker lever (with engine brake integrated when applicable) (1)
- Intake rocker lever (2).

Exhaust rocker lever is **always** the shortest rocker lever.



⚠ CAUTION ⚠

Intake and exhaust valve lash must be set using the indicators on the camshaft gear (rear or front face). Failure to do so may result in engine damage.

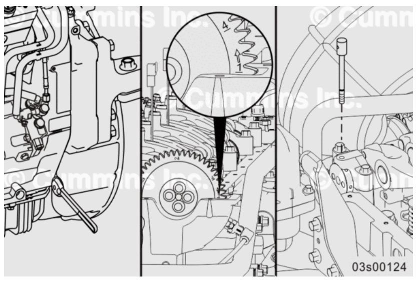


There are one bigger hole and six small holes (60 degree intervals) on the front end of camshaft for timing lock.

Camshaft gear has timing marks which are visible from the rear of the engine.

Locate the valve timing marks on the rear side of the camshaft gear.

Rotate the engine barring tool, Part Number 4919092, in the direction of engine rotation, **clockwise**. Align the timing mark 1 (bottom of the number) on the camshaft gear with the valve cover sealing flange.

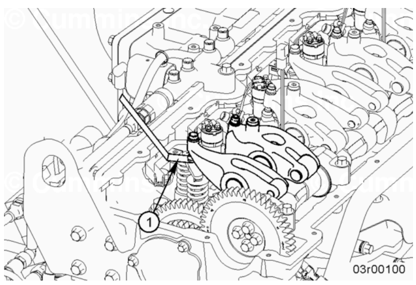


Insert camshaft timing pin, Part Number 5572844, into the small hole found on the front of camshaft to lock the camshaft. The camshaft will be locked in place when the pin is properly in place.

At timing mark 1, set the intake and exhaust valve lash for cylinder 1.

⚠ CAUTION ⚠

Engine damage can occur if the running valve lash is not within specifications.



Loosen the locknut on the rocker lever adjusting screw, and back out the adjusting screw.

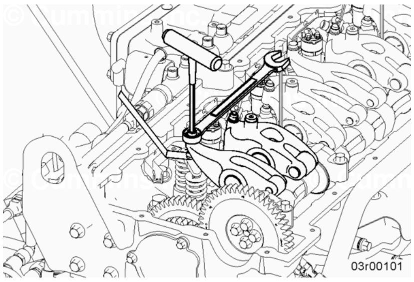
Insert the feeler gauge (1) between the bottom of the e-foot and the top of the crosshead.

Note : The valve lash must be reset to nominal specifications when maintenance interval occurs for overhead set.

Lash Reset Specifications			
	mm		in
Exhaust valve lash	0.90	NOM	0.035
Intake valve lash	0.45	NOM	0.018

Note : For engines equipped with engine brakes, constant contact will exist between the camshaft and the exhaust camshaft follower.

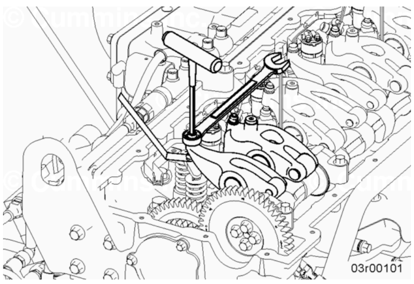
To adjust the valve lash in the intake and exhaust valves, the following steps shall be used:



1. Rotate the valve lash adjuster **clockwise** until contact is felt with feeler gauge.
2. Rotate valve lash adjuster half a revolution **clockwise**.
3. Loosen valve lash adjuster half a revolution.
4. Repeat steps 2 and 3 four times.
5. Tighten valve lash adjuster until contact is felt with feeler gauge.

Note : Do not apply torque to valve lash adjuster.

Tighten the valve lash adjuster nut holding down the valve lash adjuster.



Torque Value: 28 n•m [248 in-lb]

After tightening the locknut to the correct torque value, check to make sure the feeler gauge will slide backward and forward between the crosshead and the rocker lever with **only** a slight drag.

Remove the feeler gauge.

Remove the small timing locking pin if installed.

Repeat process to adjust all rocker levers according to Rocker Lever Adjustment Sequence chart below. **Always** start at timing mark 1 and follow firing order described in chart.

Entire overhead assembly **must** be reset every time any overhead component is removed.

Rocker Lever Adjustment Sequence		
Camshaft Timing Mark	**Set Exhaust Number**	**Set Intake Number**
1	1	1
5	5	5
3	3	3
6	6	6
2	2	2
4	4	4
Firing order 1-5-3-6-2-4		

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/1016/1016-003-011-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.